Automatic Identification System (AIS)

Introduction to Data Science with R and RStudio

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GitLab Repository
Dashboard

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Abstract

This report provides an overview of the Automatic Identification System (AIS) and the structure of the underlying vessel data. After introducing the purpose and collection of AIS data, the report examines both static and dynamic vessel information. Based on the available AIS database, different querying and sampling strategies are applied to explore vessel movements and traffic patterns. The report further includes visual analyses of AIS observations, individual vessel behaviour, and spatial traffic density. Finally, a static dashboard and an interactive Shiny application are implemented using Docker and NGINX.

1. Introduction

The Automatic Identification System (AIS) is a maritime communication system that allows vessels to automatically transmit and receive information about their identity, position, and movement [1]. Although AIS was initially introduced to improve navigational safety and support collision avoidance at sea, it has since become an important tool for maritime monitoring, fleet management, and maritime research [2]. Due to its importance for maritime safety and traffic monitoring, AIS carriage is mandatory under SOLAS (the International Convention for the Safety of Life at Sea) for certain vessel categories. This regulation requires AIS on all ships of 300 gross tonnage and above engaged on international voyages, cargo ships of 500 gross tonnage and above not engaged on international voyages, and passenger ships regardless of size [3]. AIS transponders are divided into Class A, mandatory for large vessels under SOLAS, and Class B, intended for smaller craft with limited functionality [1].

AIS data is collected by networks of receivers that capture Very High Frequency (VHF) radio waves transmitted by vessels. As VHF radio signals are limited by distance, obstacles, and the Earth's curvature, AIS coverage depends strongly on the location and height of the receiving antenna. In conventional ship-to-ship communication, AIS signals typically cover about 20 nautical miles, while shore-based AIS stations can extend this range to around 40 nautical miles. However, when AIS receivers are placed on satellites in low Earth orbit, the coverage range can increase significantly, reaching more than 1,000 nautical miles. This makes satellite-based AIS particularly useful for long-range vessel identification and tracking, especially in remote ocean areas where shore-based AIS stations cannot provide coverage [4]. Despite this advantage, satellite-based AIS is inherently more vulnerable to message collisions than terrestrial AIS. While terrestrial AIS typically operates effectively in coastal waters due to its limited reception range and localized time-slot coordination, satellites cover much larger areas and can receive transmissions from hundreds of vessels simultaneously [4, 5]. As a result, multiple AIS messages may overlap in time and frequency, which can reduce decoding performance and lead to lost position reports [5].

In terms of transmitted information, AIS data are commonly divided into static, dynamic, and voyage-related information [6]. Static information includes vessel-specific identifiers and characteristics, such as the Maritime Mobile Service Identity (MMSI), call sign, vessel name, IMO number, dimensions, and ship type [1]. This data is usually configured during installation or system setup and transmitted every six minutes or upon request [6]. Dynamic information generally includes movement- and navigation-related data, such as the vessel's position, timestamp, course, speed, and heading [1]. Most dynamic data, such as position, timestamp, speed, and course, are automatically updated by onboard systems, whereas navigational status is manually entered by the crew. The transmission frequency of dynamic information depends on the vessel's speed, course alterations, and AIS equipment type. Voyage-related information, such as draught, destination, and estimated time of arrival, is also transmitted every six minutes or upon request, but it is manually entered and updated when

necessary [6]. Consequently, manually entered fields depend on regular human input and are generally more likely to be missing, incomplete, or outdated.

Taken together, the combination of dynamic movement information, static vessel characteristics, and voyage-related attributes makes AIS a highly valuable data source for the analysis of maritime traffic. Position, timestamp, speed, course, and heading allow vessel routes to be reconstructed and movement patterns to be examined in ports, waterways, and open-sea areas. When these dynamic records are linked with static vessel information, such as vessel type, flag, or dimensions, traffic patterns can also be compared across different vessel groups. However, the analytical value of AIS data is limited by their data quality. AIS datasets are often large and affected by missing, duplicated, inaccurate, or outdated records. These issues may result from technical failures, limited receiver coverage, dense traffic, satellite message collisions, or incorrectly entered manual information. As a result, AIS data cannot be treated as a complete or error-free representation of vessel movement. They require systematic cleaning, validation, and plausibility checks before being used for maritime traffic analysis [7].

The aim of this report is to examine the analytical potential and limitations of AIS data for the study of vessel movement. AIS data provide detailed information about vessels, their characteristics, and their movement over time, but the size and quality of these datasets also create challenges for analysis. Therefore, the report considers how AIS data are structured, how large amounts of dynamic position data can be handled through sampling, and how spatial and temporal information can be used to study vessel behaviour. By using maps, plots, and dashboard elements, the project presents AIS-based traffic patterns in a clear way while also showing that the results must be interpreted with attention to possible data quality issues.

2. Data

The AIS vessel data was provided by Dr. Bleher via a PostgreSQL database, accessed through the PostgREST root endpoint at https://aidaho-edu.uni-hohenheim.de/aisdb. The API exposes ten queryable resources covering AIS records from January 1, 2021 to January 25, 2024.

Table 1: Available PostgREST API resources. The ten data tables and views exposed by the AIS database endpoint.

Endpoint	Description
`ais_static`	Static vessel information, one row per vessel
`ais_dynamic`	Dynamic vessel information
`ais_germany`	AIS records located within Germany
`geometry_columns`	Metadata on geometry columns in the database
`germany_h3`	H3 grid information for Germany
`ais_dynamic_daily_counts`	Daily counts of dynamic AIS records
`geography_columns`	Metadata on geography columns in the database
`ais_dynamic_daily_counts_mat`	Materialized daily counts of AIS records
`spatial_ref_sys`	Spatial reference system definitions
`ingested_files`	Information on imported source files

2.1 Data Sources and Structure

The AIS database consists of two central tables: `ais_static` and `ais_dynamic`. The `ais_static` table contains static vessel data, with one row per vessel identified by its MMSI. It contains 228,427 rows, representing unique vessel identifiers, and stores 13 columns. By contrast, the `ais_dynamic` table contains dynamic vessel data and therefore multiple observations per vessel over time. Due to the high-frequency transmission of AIS signals, the table contains approximately 826.79 million rows across 16 columns. Both tables are linked through the variable mmsi, which uniquely identifies vessels. Table 2 and Table 3 provide an overview of the columns contained in `ais_static` and `ais_dynamic` along with their meanings. Beside these two central tables, the database also provides the table `ais_germany`, which shares the same structure as `ais_dynamic` but is restricted to vessels tracked within Germany, and later used for a German river traffic analysis.

2.2 Limitations

Although the introduction already outlined general limitations of AIS data, they are worth reconsidering for the database used here, as they directly affect the completeness and reliability of the analysis.

Coverage must be considered in two ways. First, AIS reception depends on the receiver infrastructure: terrestrial reception is strongest near coasts, ports, and inland waterways, while satellite AIS covers much larger areas but suffers message collisions in dense-traffic zones, where overlapping transmissions are not reliably decoded [4, 5]. Second, this creates availability bias — the recorded observations are not a random draw from all transmissions but reflect what the infrastructure was most likely to capture [8]. Coastal cargo traffic, ferries,

Table 2: Columns of the `ais_static` table. Field names, descriptions, and data types of the static, one-row-per-vessel metadata.

Column Name	Description	Type
`mmsi`	Maritime Mobile Service Identity (uniquely identifying a vessel)	`INTEGER`
`imo`	International Maritime Organization number	`INTEGER`
`name`	Vessel name	`TEXT`
`call_sign`	Radio call sign	`TEXT`
`flag`	Country flag	`TEXT`
`draught`	Vessel draught in meters	`REAL`
`ship_type_code`	Numeric AIS ship type classification code	`SMALLINT`
`ship_type`	Descriptive ship type category	`TEXT`
`length`	Length in meters	`REAL`
`width`	Width in meters	`REAL`
`eta`	Estimated time of arrival reported by the vessel	`TIMESTAMP WITH TIME ZONE`
`destination`	Destination reported by the vessel	`TEXT`
`static_updated_at`	Timestamp of last static information update	`TIMESTAMP WITH TIME ZONE`

Table 3: Columns of the `ais_dynamic` table. Field names, descriptions, and data types of the dynamic position messages.

Column Name	Description	Type
`created_at`	Timestamp when the record was ingested	`TIMESTAMP WITH TIME ZONE`
`msg_timestamp`	Timestamp of the original AIS message	`TIMESTAMP WITH TIME ZONE`
`position_updated_at`	Timestamp when the vessel position was last updated	`TIMESTAMP WITH TIME ZONE`
`mmsi`	Maritime Mobile Service Identity (uniquely identifying a vessel)	`INTEGER`
`latitude`	Geographic latitude in decimal degrees	`DOUBLE PRECISION`
`longitude`	Geographic longitude in decimal degrees	`DOUBLE PRECISION`
`speed`	Speed over ground (SOG) in knots	`REAL`
`course`	Course over ground (COG) in degrees	`REAL`
`heading`	Heading in degrees	`SMALLINT`
`status`	Navigational status code reported by the vessel	`SMALLINT`
`maneuver`	Maneuver indicator reported by the vessel	`SMALLINT`
`accuracy`	Accuracy of the reported position	`BOOLEAN`
`rot`	Rate of turn (ROT)	`SMALLINT`
`collection_type`	Source of data collection	`TEXT`
`geom`	Geographic point geometry of the position	`PUBLIC.GEOGRAPHY(POINT,4326)`
`h3_index`	H3 grid index assigned to the vessel position	`BIGINT`

and port approaches are therefore over-represented, while offshore fishing, remote routes, and polar regions appear less often or with larger gaps. The database should thus be read as a coverage-dependent sample, not a complete or unbiased record of all vessel movement.

AIS data may also contain erroneous records. Raw data include invalid, abnormal, or duplicated values ("dirty" data) caused by interference, equipment faults, or device manipulation [9], which can produce incorrect positions, artificial jumps, or implausible speeds [10]. A further problem is identifier reuse: because tracks are reconstructed from the MMSI, a shared MMSI makes a track jump between locations in a characteristic zigzag [9, 10]. Unusual routes or extreme speeds should therefore be interpreted with caution, as they may reflect data errors rather than real behaviour.

Finally, the dataset is far too large to process in full, so the analysis necessarily relies on filtered extracts and samples [11]. The results therefore depend not only on the underlying data but also on the specific queries, sampling strategy, and extraction criteria used.

These limitations shape the interpretation of the results. The following analyses therefore need to be understood as being based on selected, coverage-dependent records that may contain gaps, errors, and biases caused by both the data collection process and the practical limits of working with a large shared database.

3. Descriptive Overview

Before analysing vessel movements, it is important to examine the structure and quality of the AIS static information. Several patterns are visible in the numeric variables, as shown in Table 4. The minimum of 0 across

all variables is physically not possible, suggesting that some entries use 0 as a placeholder for missing information. In addition, the mean exceeds the median for all three variables, indicating right-skewed distributions. This is especially visible for length, where the maximum value of 1,022 meters is physically implausible and suggests the presence of erroneous outliers. While length and width are relatively complete, draught is missing for approximately 47% of the observations, limiting its usefulness for further analysis.

Table 4: Numeric summary of `ais_static`. Minimum, mean, median, maximum, and non-missing count for the numeric vessel attributes.

Variable	Min	Mean	Median	Max	NonMissing
draught	0	4.7158	4	25.5	121121
length	0	63.5805	32	1022.0	216537
width	0	11.5478	8	126.0	216537

The categorical variables also show strong concentration and varying levels of reliability. As shown in Table 5, the dataset includes a wide range of ship types, but the distribution is clearly dominated by a few categories, particularly Cargo vessels (25.1%) and Fishing vessels (16.8%). This dominance is likely related to terrestrial receiver coverage near coasts and ports, combined with SOLAS carriage requirements that mandate AIS for larger cargo vessels [3]. At the same time, the category Other accounts for the largest share of observations (28.1%), showing the diversity of ship types in the dataset that occur too infrequently to be represented as separate classes. In addition, a substantial number of records contain missing ship type information (n = 21,842), corresponding to approximately 9.6% of all observations. This should be considered when interpreting the results, as missing values may limit the reliability of conclusions based on ship type.

Table 5: Ship types in `ais_static`. Frequency and share of each vessel type; percentages exclude the 21,842 records with no reported type.

Ship Type	Frequency	Percentage (%)
Other	57,974	28.1
Cargo	51,781	25.1
Fishing Vessel	34,736	16.8
Tanker	16,496	8.0
Tug	13,257	6.4
Pleasure Craft	9,014	4.4
Special Craft	7,624	3.7
Passenger Ship	7,147	3.5
Sailing Vessel	5,164	2.5
High-Speed Craft	1,227	0.6
Search and Rescue	1,201	0.6
Wing In Ground	644	0.3
Reserved	320	0.2
Percentages are calculated excluding NA values (n = 21,842)		

Although the dataset contains vessels operating under 229 distinct national flags (a further 22 vessels report no flag), the distribution is highly concentrated. As shown in Table 6, Chinese-flagged vessels alone account for almost 30% of all vessels — nearly five times the share of the next most common flag (the United States, 6.0%). Consistent with this pattern, the most common destinations are located in China. However, the destination variable is not consistently usable because over 50% of the entries are either missing or appear to be erroneous. Many records contain only placeholders, such as repeated "@" symbols, while others contain broad route descriptions rather than standardised destination names, for example "FOR ORDERS". This is likely because destination information is manually entered and therefore likely to contain inconsistent formatting, missing values, and input errors.

3.1 Targeted Filter Queries

Beyond these aggregate overviews, the PostgREST interface allows specific subsets of the data to be retrieved by combining filters in the request URL. Filters are AND-combined, and counts were obtained with the same exact-count method used for the table sizes (a `HEAD` request with `Prefer: count=exact`), so no row data has to be transferred. Table 7 reports several such queries together with the filter logic used.

Table 6: Most common flags in `ais_static`. The five most frequent national flags and their share of all 228,427 vessels.

Flag	Frequency	Percentage (%)
CN	67,214	29.4
US	13,771	6.0
PA	8,024	3.5
NL	7,635	3.3
NO	7,303	3.2
Five most common of 229 flags; percentages are of all 228,427 vessels		

Table 7: Targeted filter queries. Results of combined PostgREST filters against the AIS tables, with the filter logic used.

Query	PostgREST filter	Result
Benelux-flagged vessels	`flag=in.(BE,NL,LU)`	9,476
German cargo vessels	`flag=eq.DE` & `ship_type=eq.Cargo`	291
of those longer than 150 m	`...& length=gt.150`	76
Name contains "EXPRESS"	`name=ilike.*EXPRESS*`	572
Dynamic records, first 5 min of 2022	`msg_timestamp` in 5-min window	1,506
distinct vessels therein	distinct `mmsi`	1,100
Faster than 12 kn (2021-05-04, 14:00–14:30)	`speed=gt.12`	76
of those cargo	cross-referenced with `ais_static`	31

In total, 9,476 vessels sail under a Benelux flag, and 291 German-flagged cargo vessels are recorded, of which 76 exceed 150 m in length. A case-insensitive name search returns 572 vessels whose name contains "EXPRESS". On the dynamic side, the first five minutes of 2022 contain 1,506 position records from 1,100 distinct vessels, and within the 30-minute window on 2021-05-04, 76 vessels moved faster than 12 knots, of which 31 are cargo vessels. The number of returned rows is itself controlled with the `limit` parameter (for example `limit=5`), which is used throughout to keep responses within the API's size limit.

4. Sampling Strategy

Because the full AIS dynamic table contains several hundred million records, it is not realistic to analyse the complete dataset within this project. For this reason, a sample had to be selected.

A sampling strategy is a method for selecting a smaller representative sample from a larger dataset. Sampling is often necessary because full datasets can be too large, costly, and slow to process. However, sampling is not simply the act of using less data. A sample should preserve the relevant structure of the population so that analysis on the sample leads to conclusions that remain meaningful for the larger dataset [12]. Good sampling should therefore be:

- `Representative`: Sample reflects the dataset it was drawn from.
- `Unbiased`: No subgroup is over- or under-represented due to the selection mechanism.
- `Reproducible`: Same sample can be reconstructed.
- `Appropriately sized`: Large enough to draw reliable conclusions.

A sample is representative if the statistical feature of the variables, such as distribution, mean, correlations, rare events, etc., are similar to those of the entire dataset. Since the AIS API contains a very large amount of data, sampling is essential. However, AIS samples can become non-representative in several ways. The important dimensions of possible non-representativeness are:

- `Geographical bias`: The sample could mostly contain data around high-traffic areas, under representing data from the open ocean or less-frequent areas.
- `Temporal bias`: The sample could be only from certain days, seasons, or hours may miss daily, weekly, or seasonal traffic patterns. Furthermore, sampling by taking every n-th ping can disrupt the vessel's continuous path, reducing accuracy of speed, acceleration, etc.

- `Vessel bias`: The sample could be biased by vessel type or behaviour. For example, large commercial vessels may be over-represented because they transmit regularly and travel through monitored shipping lanes, while small fishing vessels or vessels with irregular transmissions may be under-represented.

In addition to these sampling biases, the previously mentioned availability bias must be considered again in the context of the sampling strategy [8]. AIS data is not collected uniformly from all vessels and regions, but depend on the underlying collection infrastructure [4,5]. Therefore, the available database already represents a coverage-dependent subset of all possible AIS transmissions. If a sample is drawn directly from this database without considering variables such as region, time period, vessel type, or collection type, the sample may reproduce or even strengthen this availability bias. As a result, the sampling strategy should not only reduce the dataset size, but also check whether important spatial, temporal, and vessel-related structures remain represented.

To address these challenges, two sampling strategies were implemented and compared to select 10,000 AIS observations from the millions of records available for January 24, 2024.

4.1 Interval-Based Sample

The first approach was cluster sampling, which involves dividing the population into groups (clusters) and randomly selecting a subset of these groups for sampling [12]. So each 5-minute interval on January 24, 2024, was treated as a cluster. A total of 100 intervals were randomly selected, and up to 100 AIS observations were extracted from each interval until a sample size of 10,000 observations was reached. But this strategy may introduce several biases. First, it is vulnerable to the temporal bias, as the randomly selected intervals may not fully capture traffic patterns throughout the day. As a result, periods with unusually high or low vessel activity could be over- or under-represented in the sample. Secondly, geographical bias may occur because vessel activity is not evenly distributed across space. Observations from busy ports and major shipping routes may be sampled more often than observations from remote offshore areas, reducing the representativeness of the sample.

4.2 Stratified Sampling

Given the potential biases associated with cluster sampling, a stratified sampling approach was used next. Unlike cluster sampling, stratified sampling divides the population into subgroups (strata) based on shared traits and draws observations from each subgroup in proportion to its size in the population [8]. The data were divided into strata based on ship type and hour of day. Observations were then allocated proportionally across strata and retrieved separately, ensuring that larger groups contributed more records while smaller groups remained represented. The corresponding proportional allocation is reported in Table 8. Because PostgREST offers no true random-row operator, randomness within each stratum was approximated in two stages: vessels were drawn at random from the stratum and their records were then randomly subsampled in R to the allocated count, with a fixed seed (`set.seed(2026)`) keeping the draw reproducible.

Table 8: Stratified allocation of 10,000 observations. Records on 24 January 2024 and the resulting proportional allocation, aggregated by ship type.

Ship type	Records (Jan 24)	Allocation
Cargo	21,036,091	3,305
Tanker	10,688,122	1,686
Other	8,879,565	1,401
Fishing Vessel	6,197,875	979
Tug	5,698,967	900
Passenger Ship	2,796,149	440
Special Craft	2,766,002	438
(not reported)	1,845,297	292
Pleasure Craft	1,512,689	238
Sailing Vessel	727,607	117
High-Speed Craft	589,924	95
Wing In Ground	234,093	43
Search and Rescue	234,824	42
Reserved	130,036	24

4.3 Comparison

To evaluate the representativeness of both samples, the distributions of ship type, speed, and collection type were compared.

Regarding ship type, the stratified sample contains a substantially larger proportion of commercial vessel types such as cargo ships and tankers, while the interval sample contains relatively more smaller or specialised

vessels, as shown in Figure 1. This is consistent with the temporal bias described above. By randomly selecting time windows, the interval sample may have captured periods of atypical vessel activity, leading to an over-representation of smaller and specialised vessels.

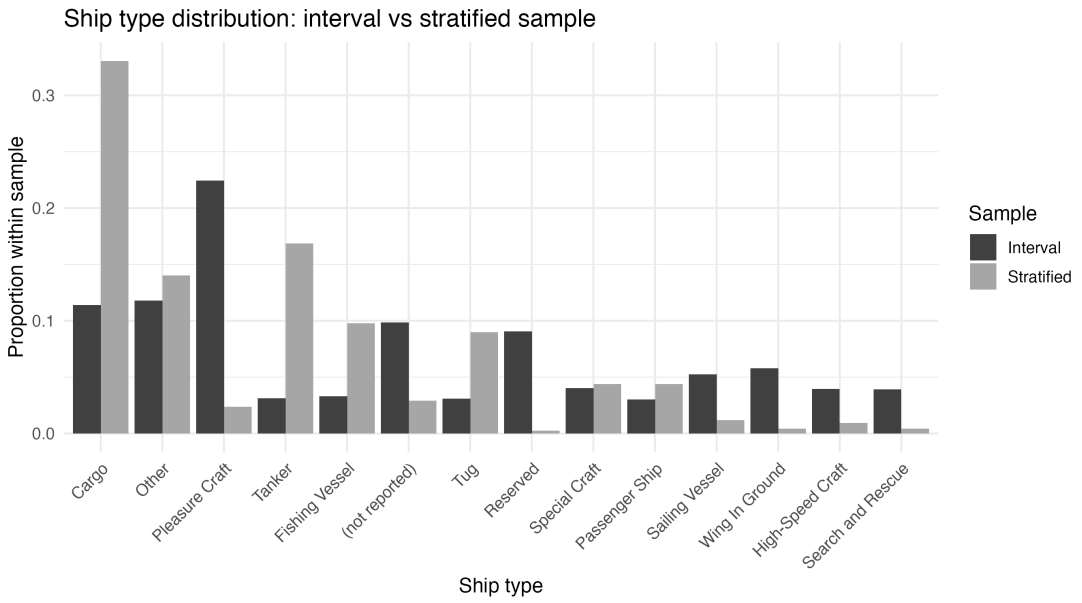


Figure 1: **Ship-type distribution by sample.** Proportion of each ship type in the interval and stratified samples.

Figure 2 compares the collection-type distribution of the interval and stratified samples. As mentioned before, Class A mainly refers to AIS transponders used by larger vessels under SOLAS, while Class B is mainly used by smaller vessels with more limited functionality [1]. Both samples are dominated by dynamic_classA. However, the stratified sample contains a lower share of dynamic_classB observations and a higher share of satellite observations. This is likely because stratified sampling reflects the overall structure of the dataset more closely, especially the strong presence of cargo ships and tankers, which are often linked to Class A and satellite-based AIS records.

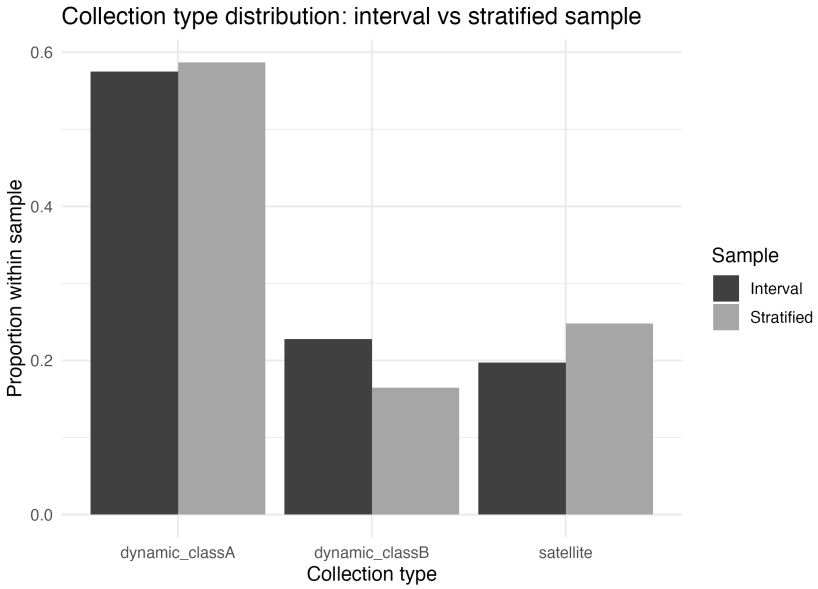


Figure 2: **Collection-type distribution by sample.** Proportion of each collection type (Class A, Class B, satellite) in the two samples.

Figure 3 makes the weakness of the interval-based sample even more clear. The interval sample is strongly concentrated at very low speeds, meaning that it includes too many stationary or very slow-moving vessels. This makes it less reliable for analysing general vessel movement, because it does not represent the full speed

structure very well. The stratified sample, on the other hand, shows a broader distribution and includes more vessels at medium speeds, especially around 10 to 12 knots. This suggests that it captures moving vessel traffic more realistically.

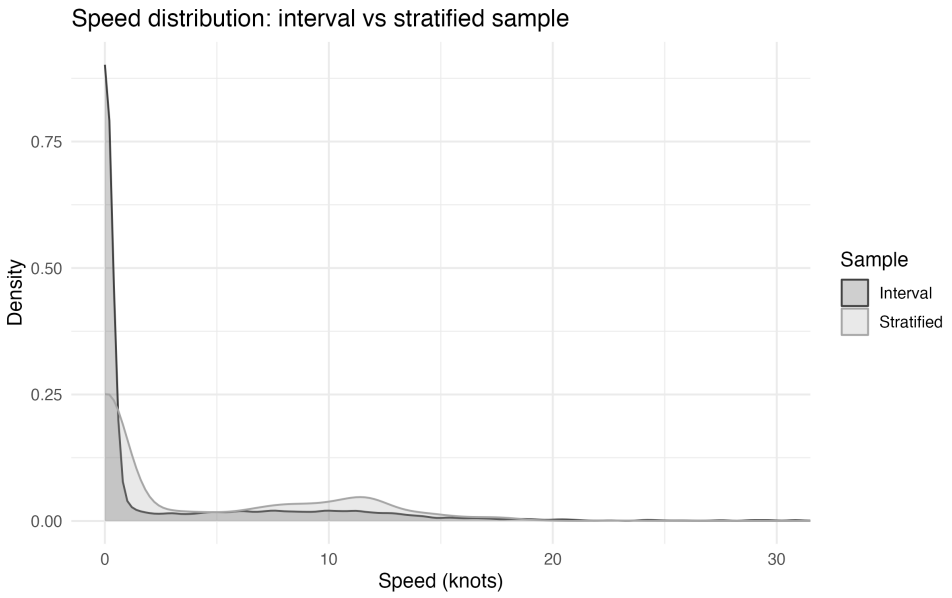


Figure 3: **Speed distribution by sample.** Kernel density of speed over ground for the interval and stratified samples.

Overall, the comparison shows that the interval-based sample is strongly affected by the randomly selected time windows. This can lead to an over-representation of specific traffic situations, such as slow-moving vessels or periods with more smaller and specialised vessels. Therefore, the interval-based sample shows a clear risk of temporal bias. The stratified sample gives a more balanced picture because it better preserves differences between ship types and hours of the day. As a result, the interval-based sample is easier to construct, but it is less representative than the stratified sample.

5. Vessel Behaviour

This section moves from aggregate statistics to the behaviour of individual vessels. It first gives a visual overview of the sampled AIS positions, then inspects three specific vessels to illustrate both data-quality problems and characteristic movement patterns, and finally develops a rule-based method for detecting ship-lock passages from a vessel track.

5.1 Sample Overview Maps

To provide a visual overview of the AIS data, Figure 4 maps all AIS positions worldwide from the previously created cluster sample. Vessel speed is represented by colour using a sequential scale capped at 25 knots, with pale yellow indicating lower speeds and dark red indicating higher speeds; the cap prevents a few implausibly high speeds from compressing the colour scale. Most observations are concentrated along major shipping routes and coastal regions, particularly around Europe, East Asia, North America, and the Mediterranean Sea.

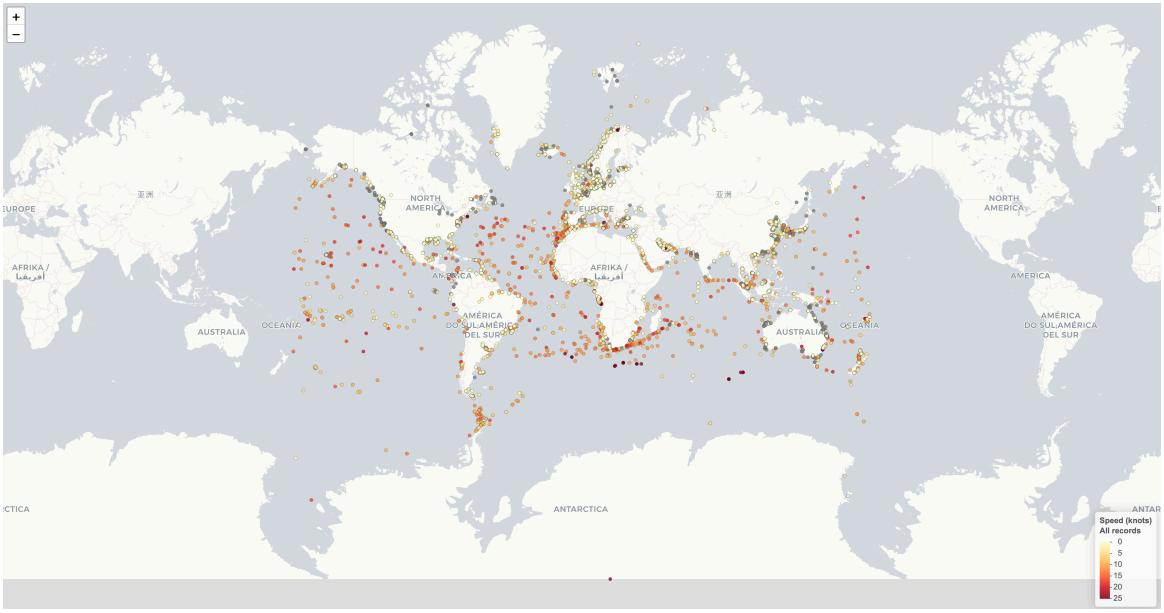


Figure 4: **Worldwide sample positions by speed.** All sampled AIS positions, coloured by speed over ground (knots, capped at 25).

A second map (Figure 5) was generated using only observations collected via satellite AIS. Compared with the full sample, the satellite-only map appears less dense and contains noticeably fewer observations in coastal areas. This pattern is expected: near coasts and busy ports, large numbers of vessels transmit simultaneously, so satellites receive many overlapping messages and signal collisions reduce the number of successfully decoded reports, whereas terrestrial receivers cover exactly these dense coastal zones well [5]. The satellite layer therefore contributes most in open-ocean areas that lie beyond terrestrial range.

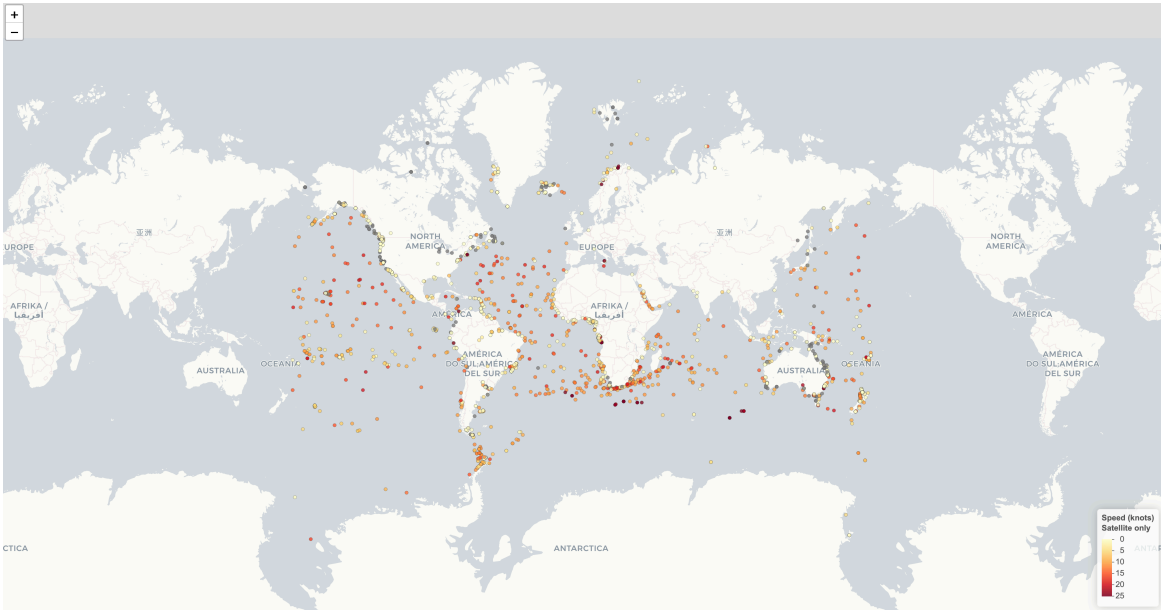


Figure 5: **Satellite-only sample positions.** The sampled AIS positions collected via satellite AIS, sparser near coasts than the full sample.

5.2 Individual Vessel Tracks

MMSI 2579999 — a data-quality example. Querying all records for MMSI 2579999 returns 10,828 messages within a single day, but the data are clearly not those of one physical vessel. The speed field is missing for every record, and the 7,827 distinct positions are scattered across the entire globe, spanning latitudes from about $-38°$ to $65°$ and longitudes from about $-58°$ to $148°$. Consecutive messages, only seconds apart, jump between the North Sea, south-eastern Australia, and the coast of South Korea, which implies physically impossible speeds. Such a pattern is a textbook symptom of a shared or default MMSI: the same identifier is transmitted by many unrelated devices, or used as a placeholder, so that reconstructing a single track produces

the characteristic worldwide "zigzag" [9,10]. The identifier should therefore be treated as unreliable and excluded from movement analysis rather than interpreted as a real voyage.

MMSI 563040400 — a vessel at berth. This Singapore-flagged vessel (MMSI prefix 563) reports 4,313 records over roughly four days (1–5 January 2021). All positions fall within a very small area around 44.0°–44.1°N and 28.7°–28.8°E, which corresponds to the Port of Constanţa on the Romanian Black Sea coast, and the speed is zero for the large majority of records (median 0 knots, maximum 11.7 knots). The vessel is thus essentially stationary at berth for the observed period, with only occasional slow movements that likely reflect repositioning within the port or minor GPS noise. Figure 6 shows the corresponding track.

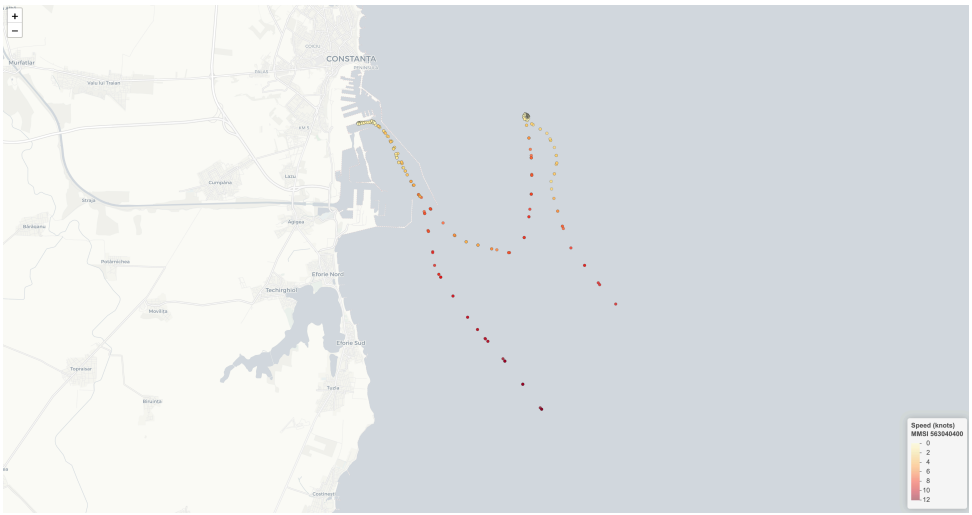


Figure 6: **Track of MMSI 563040400 (Port of Constanţa).** The vessel's positions over about four days, coloured by speed (knots, capped at 12).

5.3 Ship-Lock Detection

The vessel with MMSI 211430830 (a German-flagged vessel) was analysed over the window 2021-01-17 to 2021-01-20. The retrieved track contains 574 records between 18 and 19 January and runs along the Danube from the area east of Vienna (≈ 17.7°E) upstream to Linz (≈ 14.3°E), at a typical river speed of around 3–4 knots (median 3.6 knots).

Along this route the vessel passes several ship locks, where vessels are raised or lowered to bypass dams. *In AIS data, a lock passage has a characteristic signature*: the vessel's speed drops to near zero, its position stays within a very small area (the lock chamber, only tens of metres across) while the gates close and the water level changes, and because the vessel lingers, a comparatively high number of AIS messages accumulate in that small location. The passage typically lasts in the order of 15–30 minutes. Between locks, the vessel moves continuously at a few knots. In this track, 79 of the 574 records have a speed below 1 knot, consistent with several such stops.

This signature motivates a transparent, rule-based detection heuristic (`lock_detection.R`). The decision logic is:

1. Sort the track chronologically by `msg_timestamp`.
2. Keep only *near-stationary* points with speed below 1 knot as candidate lock points.
3. Group consecutive candidates together, starting a new group whenever the time gap to the previous candidate exceeds 30 minutes *or* the spatial jump exceeds 300 metres (great-circle / haversine distance).
4. Discard groups that are too short to be a plausible lock passage (shorter than 5 minutes or fewer than 3 observations).
5. For each remaining group, report the start time, end time, duration, mean position, and number of observations.

The thresholds reflect the physical process: 1 knot separates genuine stops from normal river travel; 300 metres keeps a group inside a single lock chamber while tolerating GPS scatter; and the 5-minute / 3-point minimum removes brief drifts and isolated noise. The 30-minute gap allows for sparse reporting while a vessel waits.

Applying the heuristic to the track yields the five candidate passages in Table 9, also shown on the map in Figure 7, where detected locks are marked in red along the blue trajectory. Four of the five are plausible lock passages of 18–25 minutes spaced along the Danube between Vienna and Linz. The fifth, lasting 132 minutes

at the western end of the route (14.35°E, Linz), is almost certainly the vessel's final mooring at its destination rather than a lock; it is flagged because a long stop at one location matches the same low-speed, tightly clustered pattern.

Table 9: Detected ship-lock passages for MMSI 211430830. Start, end, duration, mean position, and observation count of each candidate lock on the Danube (18–19 January 2021).

Start (UTC)	End (UTC)	Dur. (min)	Lat	Lon	Obs
2021-01-18 05:21	2021-01-18 05:46	24.5	47.883	17.541	6
2021-01-18 19:20	2021-01-18 19:41	21.7	48.176	16.479	8
2021-01-18 20:55	2021-01-18 21:18	23.0	48.244	16.386	12
2021-01-19 20:45	2021-01-19 21:03	18.2	48.250	14.432	7
2021-01-19 21:45	2021-01-19 23:57	132.6	48.278	14.349	45

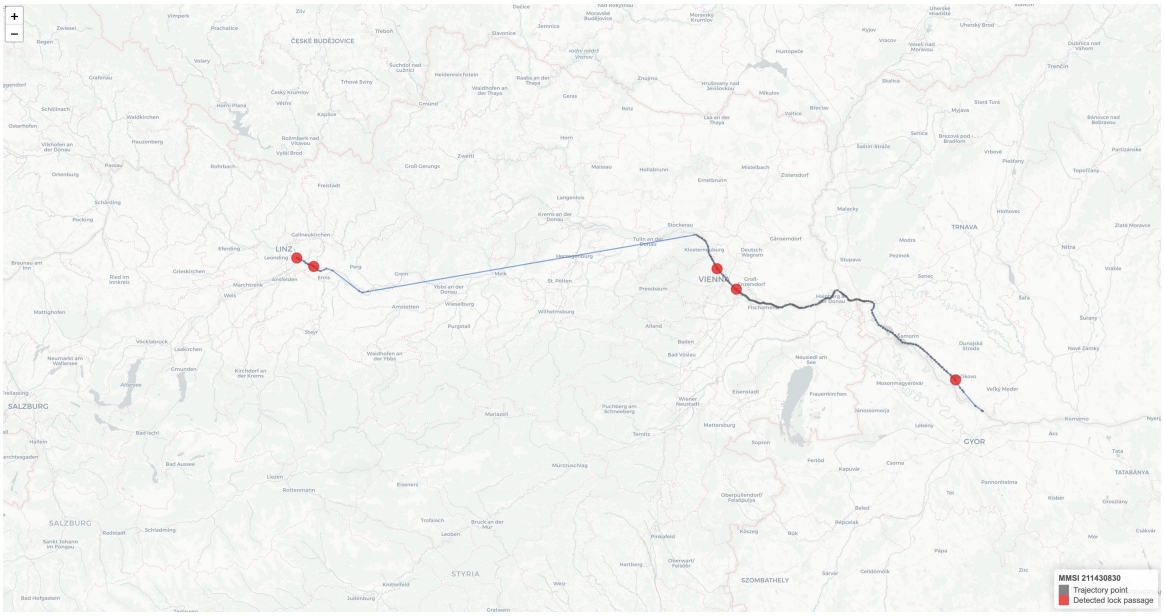


Figure 7: **Lock passages of MMSI 211430830.** The Danube trajectory (blue) with detected ship-lock passages highlighted in red.

The goal of this heuristic is transparency and reproducibility rather than perfect recall. Its limitations follow directly from the rules: a long stop that is not a lock (such as the final mooring above) is reported as a false positive, while a lock taken without dropping below 1 knot, or with very sparse reporting, could be missed. The fixed thresholds are also tuned to inland navigation and would need adjustment for other vessel classes. A natural improvement would replace the manual thresholds with an *unsupervised* approach: rather than hand-labelled training data, a density-based clustering method such as DBSCAN could be applied to the low-speed points in space and time, letting the data define the clusters and adapting automatically to different reporting rates and lock geometries.

6. AIS Traffic Density on German Rivers

Building on the dynamic AIS records, this section combines vessel positions with geographic information about German rivers to analyse how traffic is distributed across the inland waterway network. The analysis is based on the `ais_germany` table, which contains AIS records for vessels tracked within Germany, and uses the H3 spatial indexing system to assign each record to a river.

6.1 River Geometries and Study Data

River geometries for Germany were obtained from the `rnaturalearth` package, which provides global river centre-lines as line geometries. The raw layer required cleaning before it could be used. River features were first clipped to the German national boundary, and empty or unnamed fragments were removed. The same river sometimes appears under different language spellings; the German, English, and French forms of the Rhine (*Rhein*, *Rhine*, *Rhin*) were therefore merged into a single feature, and multi-part segments of each river were dissolved into one geometry per river. Finally, rivers with a negligible length inside Germany were dropped:

the Warta, for example, is almost entirely Polish and reduces to a sliver of about ten metres after clipping, so it does not represent meaningful German inland traffic. This cleaning leaves nine major rivers: the Danube (*Donau*), Elbe, Ems, Inn, Main, Mosel, Oder, Rhine, and Weser.

Because only nine rivers remain, each can be shown in its own colour with a fully readable legend, so no further reduction to a subset of major rivers was necessary. Figure 8 maps the cleaned geometries over the German boundary.

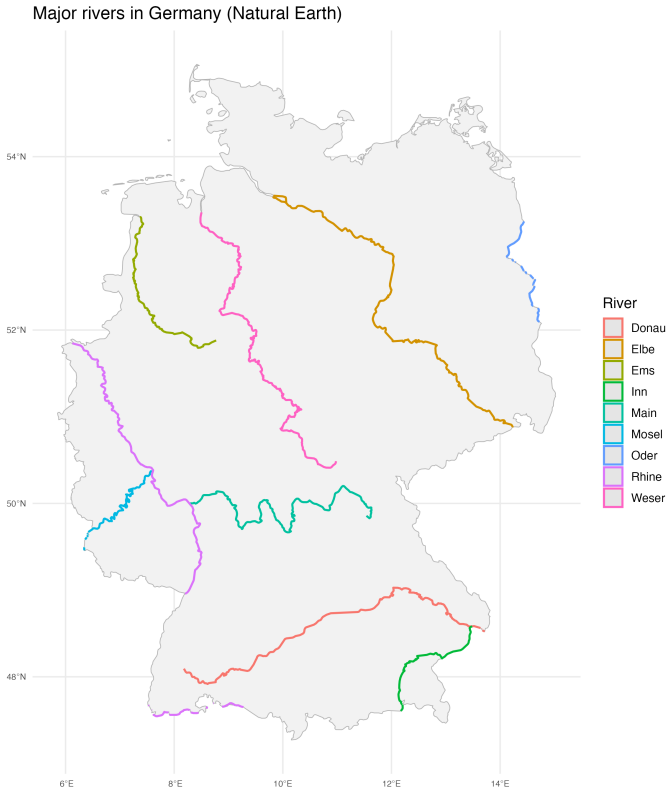


Figure 8: **German river geometries.** The nine cleaned rivers, each shown in a distinct colour (Natural Earth).

The `ais_germany` table was then queried for all tracking records on 2022-04-23, using a timestamp filter that returns only records with a message timestamp on that date. This yields 497,990 AIS records, which form the basis for the spatial matching below.

6.2 Spatial Matching with H3

Determining which records lie on a river is a spatial-join problem. A naive solution would test every AIS point against every river geometry, either by point-in-polygon checks or by computing the distance from each point to every river segment. With roughly half a million points and nine multi-part geometries, this scales as the product of the two and becomes computationally expensive. The H3 indexing system avoids this cost. H3 divides the world into a hierarchical grid of hexagonal cells, each identified by a unique integer index, with the resolution controlling cell size (coarse continent-scale cells at low resolutions, fine metre-scale cells at high ones) [13]. Converting a coordinate to its H3 index reduces the spatial-join to a key lookup: each cell index acts as a key in a hash table, a structure that maps keys to values with average constant-time retrieval. Instead of comparing geometries, the analysis only has to check whether a point's cell index appears in the set of cell indices covered by the rivers, which turns an expensive geometric comparison into a fast set membership test that scales linearly with the number of points. Hexagonal cells additionally have uniform adjacency, since every neighbour shares an edge at an equal centre-to-centre distance, which makes them better suited to distance-based reasoning than square grids.

Following the task specification, H3 resolution 7 was used, where a cell spans roughly 1.2 km. Each AIS record was converted to its H3 cell from its latitude and longitude. The rivers, however, are line geometries and cannot be filled with cells directly, so each river line was buffered into a thin corridor before being converted to cells. A buffer of 750 m (a corridor about 1.5 km wide) was chosen because it is slightly wider than one resolution-7 cell: this keeps each river at least one cell wide along its whole length and absorbs the offset between a vessel's reported position and the mapped river centre-line, while remaining narrow enough not to absorb large amounts of nearby non-river traffic. Buffering was performed in a metric projection (UTM 32N) so that the

distance is expressed in true metres. This produced 1,168 unique river cells in total.

Matching the records to rivers reduces to checking each record's cell against this set. Where a single cell was claimed by more than one river, typically at confluences such as the Rhine/Main or Rhine/Mosel junctions, a deterministic rule was applied: the cell is assigned to the larger river, measured by its total number of cells. Only two cells were affected, so the specific tie-break has no material effect on the results, but the rule is documented for reproducibility.

6.3 Traffic Distribution Results

Of the 497,990 records on 2022-04-23, 183,408 (36.8%) fall on a German river, while 314,582 (63.2%) were recorded elsewhere, such as coastal or open-water areas. A clear majority of German AIS traffic on this date therefore occurs outside the inland river network, which is consistent with the dominance of coastal and maritime shipping in the dataset.

The composition of traffic also differs between the two groups. Figure 9 compares the ship-type distribution on rivers and elsewhere. Off-river traffic is dominated by Cargo vessels (29.3%), followed by Passenger Ships (20.6%) and Tankers (9.2%). On rivers, the share of Cargo drops to 19.0% while Passenger Ships make up a comparable 19.4%, reflecting the role of ferries and excursion vessels on inland waterways. Records without a reported ship type are the single largest group in both cases and are especially prominent on rivers (46.8%), which is consistent with a higher presence of smaller inland craft that more often transmit incomplete static information.

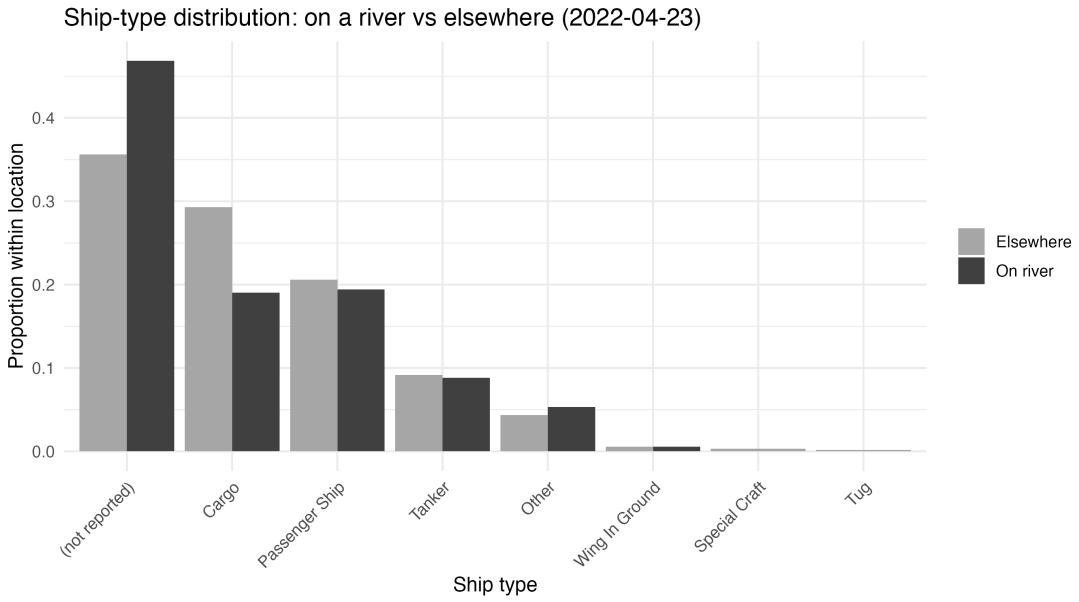


Figure 9: **Ship types on rivers vs. elsewhere.** Share of each ship type among records matched to a German river compared with those tracked elsewhere (2022-04-23).

Aggregating the on-river records by river gives the ranking in Table 10. The Rhine carries by far the most traffic with 89,033 records, followed by the Elbe (62,680) and the Weser (20,362); the remaining rivers each account for a few thousand records or fewer. The Inn and Oder, although they contribute H3 cells, returned no matched records on this date and are therefore not listed, which is plausible given their limited and largely border-bound inland AIS coverage.

Table 10: AIS records per German river (2022-04-23). Matched record count per river, in descending order.

River	AIS records
Rhine	89,033
Elbe	62,680
Weser	20,362
Donau	3,895
Main	3,768
Mosel	2,510
Ems	1,160

Finally, the distribution of traffic *along* the Rhine was examined in more detail. The 89,033 Rhine records

were projected onto the Rhine geometry to obtain each point's distance along the river, which was then binned into 10 km intervals; Figure 10 shows the resulting density. This step requires two assumptions about the reference geometry. First, the clipped German Rhine consists of several disconnected pieces, so a single continuous reference line was constructed by orienting each piece from north to south and joining the pieces in order of latitude; because the joined line bridges the gaps between pieces, the resulting distance scale (about 1,078 km) somewhat overstates the true navigable channel and should be read as a relative position along the river rather than exact kilometrage. Second, distance is measured from the northernmost point southward, so 0 km corresponds to the lower Rhine near the Dutch border and increasing values move upstream. With these assumptions, the plot shows traffic concentrated in distinct stretches rather than spread evenly, with pronounced peaks that correspond to the busier lower- and middle-Rhine reaches.

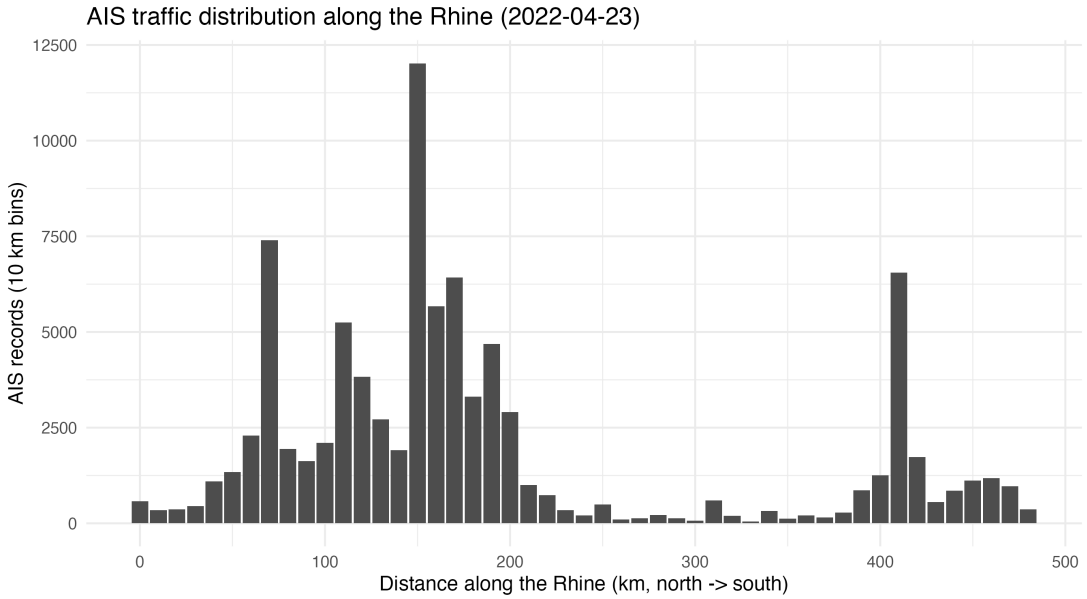


Figure 10: **Traffic distribution along the Rhine.** AIS records in 10 km bins (2022-04-23), measured north to south.

7. Dashboard and Deployment Evidence

Both web components are deployed on the assigned server through a single Docker Compose stack (an NGINX container and a Shiny container on a shared network) and are reachable behind the NGINX reverse proxy.

The **static dashboard** is the speed-coloured interval-sample map from Section 5, exported as a self-contained HTML page (`02_code/R/sample_html_dashboard.R` via `htmlwidgets::saveWidget()`) and served from the NGINX document root. A reader sees the 10,000 sampled AIS positions as circle markers coloured by speed over ground (capped at 25 knots; pale yellow = slow, dark red = fast), clustered at low zoom for legibility, with a speed legend.

The **interactive Shiny application** reuses the river traffic-density logic from Section 6. The user selects a river and one or more ship types and clicks `refresh` to update the traffic-density plot for 2022-04-23. To avoid repeated expensive API calls from the deployed app, the required data are pre-computed once `02_code/R/prepare_shiny_data.R` and written to a CSV that the app reads at start-up. NGINX exposes the app under the `/ais_app` endpoint.

Table 11: Dashboard and deployment evidence. The two deployed web components and where each can be verified.

Component	Evidence
Static map dashboard	Live at http://193.197.231.174:7080/sample_points.html; see Figure 11.
Interactive Shiny application	Live at http://193.197.231.174:7080/ais_app/; see Figure 12.

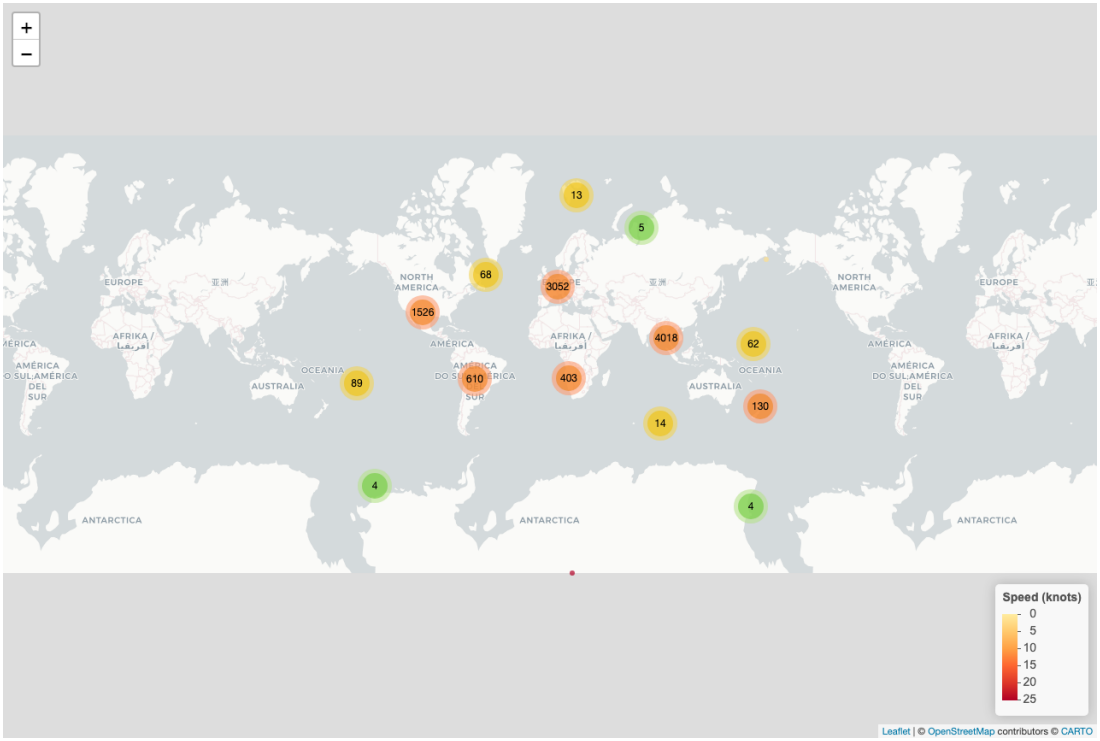


Figure 11: **Deployed static dashboard.** The speed-coloured sample map served by NGINX under `/sample_points.html`.

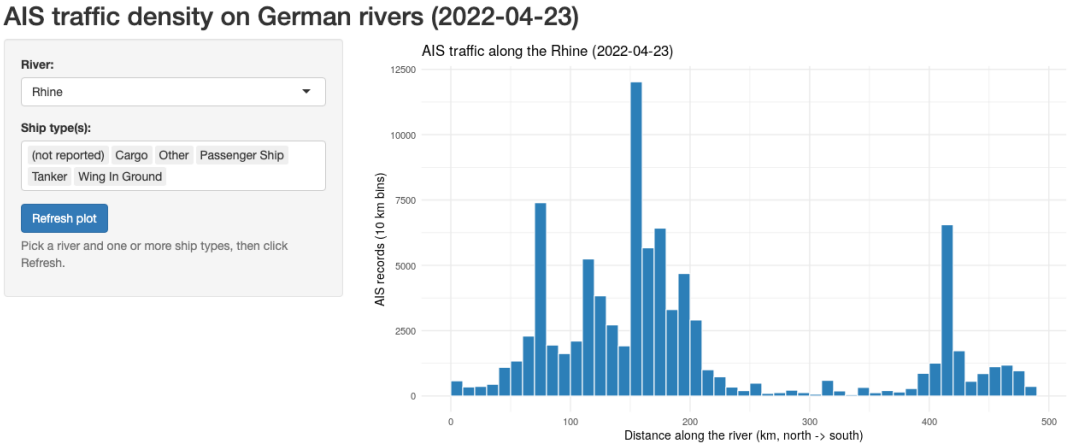


Figure 12: **Deployed Shiny application.** The interactive river traffic-density app served under `/ais_app`.

Both endpoints were confirmed to load and respond in a browser; the Shiny inputs update the plot when the `refresh` button is clicked.

8. Conclusion

This report examined the analytical potential and the limitations of AIS data for the study of vessel movement. After introducing how AIS data are generated, transmitted, and collected, the descriptive overview showed that the database is large but uneven: key fields such as draught or destination are frequently missing or contain

placeholder values, and the data are concentrated on a few ship types, flags, and regions, reflecting both SOLAS carriage rules and coverage-dependent reception.

Because the full dynamic table is far too large to analyse directly, two sampling strategies were compared. The interval (cluster) sample proved easy to construct but sensitive to the randomly chosen time windows, over-representing slow and stationary vessels, whereas the stratified sample preserved the structure of ship types and hours more faithfully and was therefore the more representative basis for analysis. The vessel-behaviour analysis then illustrated, on individual tracks, both a clear data-quality failure (a shared or default MMSI producing a physically impossible worldwide track) and characteristic real patterns (a vessel idle at berth in Constanţa, and a Danube voyage whose ship-lock passages were recovered by a simple, transparent low-speed clustering heuristic). Combining the AIS records with river geometries through H3 spatial indexing showed that about 37% of German traffic on the study date ran on the inland river network, dominated by the Rhine and the Elbe. Finally, the results were made accessible through a deployed static dashboard and an interactive Shiny application.

Overall, the project demonstrates that AIS data support rich analyses of maritime and inland traffic, but only when their gaps, biases, and errors are handled explicitly: through deliberate sampling, plausibility-aware interpretation, and transparent, reproducible processing. The same coverage and quality limitations that shape the results also mark the natural directions for future work, such as multi-day analyses, more sophisticated (for example unsupervised) event detection, and systematic data cleaning before interpretation.

9. Declaration on the Use of AI Tools

Claude (Anthropic), via the Claude Code CLI, was used as a coding and writing assistant: drafting and editing report sections, constructing and debugging the PostgREST queries and R scripts, and configuring the Docker/NGINX deployment. All code and text were reviewed, tested, and validated by us, and the analytical decisions and the interpretation of results remain our own. No AI-generated citations were used without checking the original sources.

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A Appendix: Reproducibility and Repository Information

This appendix collects the technical information needed to reproduce the results, kept here rather than in the main narrative. A full, file-by-file description of the repository and the run instructions is given in the `README.md` at the repository root.

A.1 Repository Structure

The submission follows the folder structure required in Task 1:

- `00_docs` — assignment sheet and supporting documents.
- `01_data` — generated samples and cached intermediate data products.
- `02_code/R` — analysis scripts, with reusable helpers in `functions/`.
- `03_report` — this report and, in `graphs/`, every figure it uses.

The deployment files (`docker-compose.yaml`, `nginx/`, `shiny/`, `www/`) live at the repository root and are described in the `README.md`.

A.2 Execution Order and Reproduction Notes

All scripts use paths relative to the repository root; opening `aidaho_ids_thas_2026.Rproj` in RStudio sets the working directory accordingly. Random samples use a fixed seed (`set.seed(2026)`), and large API results are cached as `.rds` files in `01_data/`, so a re-run uses the cache instead of re-querying the shared database. The per-script task mapping, and the three scripts that require live API access, are listed in the `README.md`.

Table 12: Technical reproduction overview. Main input and output of each analysis step.

Analysis step	Main input	Main output
Data overview	AIS database (PostgREST)	overview tables
Sample construction	`ais_dynamic` messages	`sample_intervals.csv`, `sample_stratified.csv`
Vessel behaviour	selected AIS tracks	vessel maps, lock-event table
River traffic	`ais_germany`, river geometries	river-traffic figures
Deployment	prepared CSV	static dashboard, Shiny app

A.3 Software Environment

Table 13: Software environment. Versions and tools used to produce the submitted results.

Component	Version or description
R version	4.6.0 (2026-04-24)
Operating system	macOS 26.5 (Apple silicon, `aarch64`)
Package management	`install.packages()`; no `renv` lockfile. The required packages and the install command are listed in the `README.md`.
Deployment	Docker Compose (NGINX + Shiny)

A.4 Important Submitted Files

- `README.md` — repository overview and run instructions.
- `01_data/sample_intervals.csv` — interval (cluster) sample (Task 3.2).
- `01_data/sample_stratified.csv` — stratified sample (Task 3.3).
- `03_report/graphs/` — all figures used in this report.
- `docker-compose.yaml`, `shiny/`, `nginx/`, `www/` — WebApp deployment (Tasks 6–7).